

Duggireddy Hitesh Pranav Reddy

+91 8296465473 | [Email](#) | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)



EDUCATION

PES University

B.Tech in Electronics and Communication Engineering (CGPA: 8.37, As of 6th Sem)

Bengaluru, India
Sept. 2023 – July 2027

Sri Chaitanya College of Education

High School Diploma (Class 12th - 91.4%, Class 10th - 83.2%)

Bengaluru, India
2011 – 2023

EXPERIENCE

Joy of Engineering Lab (JoEL), PES University

Student Head

Bengaluru, India
Aug. 2024 – Present

- Led backend planning and logistics for HackeZee 2025 and Roadshow 2026.
- Coordinated volunteers, mentors, vendors, and technical teams for engineering events.
- Mentored 6+ student teams on IoT and embedded systems projects.
- Developed and maintained the official JoEL website using Next.js and TypeScript.

C-ISFCR, PES University

Research Intern

Bengaluru, India
June 2025 – July 2025

- Developed an ML-based Intrusion Detection System for automotive CAN bus networks.
- Built a Python firewall for real-time anomaly detection and traffic filtering.
- Trained ensemble machine learning models using Scikit-learn.
- Developed a Tkinter GUI for CAN message injection, testing, and analytics.

PROJECTS

Automated Hardware Trojan Detection & ATPG Analysis *Python, Verilog, ATPG, SCOAP, DFT*

- Engineered a Python automation framework to parse gate-level Verilog netlists and programmatically insert Hardware Trojans into ISCAS-85 benchmark circuits.
- Generated comprehensive fault lists and test vectors, executing Automatic Test Pattern Generation (ATPG) using Atalanta for rigorous fault coverage analysis.
- Conducted structural graph and toggle rate analysis utilizing SCOAP testability metrics to compare clean and Trojan-inserted combinational circuits.
- Visualized suspicious gate-level activities to evaluate the effectiveness of generated test vectors in detecting malicious modifications.

Automotive ML IDS *Python, Scikit-learn, Tkinter, Machine Learning*

June 2025 – July 2025

- Built an intrusion detection system using ensemble machine learning models.
- Integrated a real-time firewall for CAN bus traffic monitoring.
- Improved anomaly detection using feature engineering and model optimization.

JoEL Website *Next.js, TypeScript, Tailwind CSS, Performance Optimization*

- Designed and developed the official JoEL website using Next.js, TypeScript, and Tailwind CSS.
- Built an interactive inventory management system to streamline equipment tracking and availability monitoring.

PESU Research Match *Flask, PyTorch, Sentence-Transformers, Docker, Qwen LLM*

- Built a hybrid semantic search engine combining dense vector retrieval (BGE-Large) and BM25 lexical search for faculty recommendation.
- Integrated a local LLM (Qwen2.5-0.5B) to generate personalized match explanations and deployed via Docker on Hugging Face Spaces.

VRE Website *React, Vite, Three.js (React Three Fiber), GSAP, Tailwind CSS*

- Migrated and revamped the official Vega Racing Electric website using Vite for high performance and fast load times.
- Integrated interactive 3D car rendering with React Three Fiber and built cinematic scroll animations using GSAP.

SKILLS

Languages : Python, C, C++, SQL, HTML, CSS, JavaScript, TypeScript, Verilog

Frameworks/Libraries : React, Next.js, FastAPI, Tailwind CSS, Scikit-learn, PyTorch, Tkinter, Chart.js, Three.js, GSAP

Core CS : Data Structures & Algorithms, Object-Oriented Programming

Tools/Platforms : Git & GitHub, Linux, Docker